

Exercise 1: Basic Observable & Subscription

Topic: Creating an Observable and subscribing to it.

Problem:

Create an Observable that emits the numbers 1, 2, 3, 4, 5 sequentially (synchronously). Subscribe to it and print each emitted value to the console. Also, handle the completion by logging "Observable completed".

Exercise 2: Using Operators – `map` and `filter`

Topic: Transforming and filtering data streams.

Problem:

Create an Observable that emits numbers from 1 to 10 using `range(1, 10)`. Use the `pipe` method with:

- `filter` to keep only even numbers.
- `map` to multiply each even number by 10.

Subscribe and log the results.

Expected Output:

20, 40, 60, 80, 100

Exercise 3: Basic Fetch Wrapped in Observable

Topic: Converting a Promise-based fetch into an Observable.

Problem:

Use the JSONPlaceholder API (free fake API) to fetch a single todo item: <https://jsonplaceholder.typicode.com/todos/1>

1. Create a function `fetchTodoAsObservable(id)` that uses `new Observable()` to wrap the `fetch` API.
2. Inside the Observable, call `fetch` and emit the parsed JSON response using `observer.next()`.
3. Handle errors with `observer.error()`.
4. Subscribe to the Observable and log the received todo object to the console.

Expected Output (example):

```
{  
  userId: 1,  
  id: 1,  
  title: "delectus aut autem",  
  completed: false  
}
```